

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVPE604F - STRUCTURAL AUDIT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Structural audits are mainly conducted to:
 - A) Identify structural defects and recommend remedial measures
 - B) Increase floor area
 - C) Improve aesthetics only
 - D) Reduce property tax
- 2 Visual inspection is the:
 - A) Method of load testing only
 - B) Last stage of demolition
 - C) First stage of a structural audit
 - D) Laboratory testing procedure
- 3 The main limitation of visual inspection is that it:
 - A) Requires expensive equipment
 - B) Cannot detect hidden internal defects
 - C) Is destructive in nature
 - D) Measures compressive strength directly
- 4 Sulphate attack on concrete primarily causes:
 - A) Expansion and cracking
 - B) Increase in strength
 - C) Improved durability
 - D) Reduction in permeability only
- 5 High permeability of concrete generally leads to:
 - A) Increased durability
 - B) Easy ingress of water and harmful chemicals
 - C) Higher compressive strength
 - D) Lower porosity

- 6 Cavitation damage is commonly observed in:
 - A) Timber roofs only
 - B) Masonry walls only
 - C) Hydraulic structures and turbines
 - D) Brick pavements only
- 7 Ultrasonic Pulse Velocity (UPV) test is used to assess:
 - A) Quality and uniformity of concrete
 - B) Brick dimensions
 - C) Soil compaction
 - D) Pavement thickness
- 8 Half-cell potential method is used for:
 - A) Measuring concrete density
 - B) Corrosion potential assessment of reinforcement
 - C) Soil permeability testing
 - D) Water quality testing
- 9 Reserve strength refers to:
 - A) Additional strength available beyond service loads
 - B) Minimum required strength only
 - C) Strength after demolition
 - D) Strength of reinforcement only
- 10 Structural strengthening is generally recommended when:
 - A) Plumbing fails
 - B) Building colour fades
 - C) Existing strength is inadequate for required loads
 - D) Electrical wiring becomes old
- 11 Structural audit is particularly important for buildings affected by:
 - A) Interior decoration only
 - B) Normal painting work
 - C) Earthquakes and fire
 - D) Landscaping only
- 12 Legal provisions related to structural audits mainly ensure:
 - A) Public safety and compliance with regulations
 - B) Increased construction cost
 - C) Faster demolition only
 - D) Reduced documentation

Q2. Solve the following.

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|---|---|
| A) Explain the objectives, importance and need for structural audit of buildings. Discuss the various stages involved in conducting a structural audit. | 6 |
| B) Explain the scope, coverage and limitations of visual inspection. Discuss the factors to be observed during the visual inspection of buildings. | 6 |

Q3. Solve the following.

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| A) Explain the causes of deterioration in reinforced concrete structures. Discuss permeability and capillary porosity. | 6 |
| B) Explain the causes of deterioration in steel structures. Discuss different types of corrosion and defects in steel connections. | 6 |

Q4. Solve Any Two of the following.

- A) Explain the principles and applications of Rebound Hammer and Ultrasonic Pulse Velocity tests for concrete strength assessment. 6
 - B) Discuss carbonation test, chloride test and corrosion potential assessment for evaluating reinforced concrete structures. 6
 - C) Explain fire damage assessment using Differential Thermal Analysis and X-ray Diffraction. 6
- Q5. Solve Any Two of the following.
- A) Explain the procedure for strength evaluation of existing reinforced concrete structures. 6
 - B) Discuss the concepts of reserve strength and identification of critical sections in structural assessment. 6
 - C) Explain the methods of evaluation of damage in reinforced concrete structures. 6
- Q6. Solve Any Two of the following.
- A) Explain the approach for conducting a structural audit. 6
 - B) Discuss the legal aspects of structural auditing and explain the responsibilities of the owner and structural auditor. 6
 - C) Explain the contents of a structural audit report. 6

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